Sonic Firmware Documentation V47

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🙏 Many thanks to Sonic Team : Zylka, Kolyan, Iggy, Toni, Yves, Francois and Adam.

This software is a fork of Armel F4HWN firmware with code from NTOIVOLA, EGZUMER and DUAL TACHYON.

EXPLANATION OF BUTTON SYMBOLS:

1 – Simple press.

L1 – Long press.

F1– Press after the F button.

MR MAIN SCREEN (CHANNELS) | VFO (FREQUENCY)

L1 – (MR) **–** Transfer the channel frequency to VFO mode.

L2 – (MR and VFO) **–** VFO change (in DUAL icon indication).

L3 – Change the MR/VFO mode on the selected channel.

L4 – (MR and VFO) **–** Bandwidth change.

L5 – (MR) **–** Change the list number for the channel. (VFO) **–** Change of step.

L6 – (MR & VFO) **–** Power Change (L/M/H).

L7 – (MR and VFO) **–** On/Off flashlight indication when receiving a signal.

L9 – (MR and VFO) **–** On/Off constant backlight.

L0 – AM/FM/USB Modulation Change

F2 – (MR) **–** Delete Channel. (VFO) **–** Preserve the frequency.

F6 – Spectrum analyzer in scanrange mode.

F7 – Spectrum analyzer in scanlist mode.

F8 – Band Spectrum Analyzer.

F9 – Spectrum analyzer in frequency mode.

F0 – Radio broadcasting (FM radio).

SK1 and **SK2**—Side keys, assignable buttons.

<= and **=>** – Navigation buttons.

MA – Menu.

EXIT – Exit.

#F – Functional "F".

OPERATION IN SPECTRUM MODES

Now the scan is interlaced: the step is minimal 25k and the spectrum is swept with different start frequencies to cover all the frequencies.

For example 125.100, 125.125,… then 125.10833,125.133333…then 125.11666,125.3666…

This to improve detection.

Different types of spectrum are called by the quick buttons

F6 - Range, Freq start, stop and step set in **[5]** menu.

Scans between 2 selected frequencies with the given step

F7 - Scanlists

Scans the frequencies in scanlists that can be defined in VFO or Chirp*.

20 lists available, maximum 500 channels to scan.

F8 - Bands

Scans ranges that are defined in bands, can be defined in Chirp* or in a channel greater than 974

Channel Frequency is START, Offset + is STOP, STEP is step.

F9 - Frequency: spectrum around VFO frequency, 128 frequencies with current step.

*See Chirp section below.

1 – Skip the frequency in the scan mode.

2 – Turn on/off the backlight).

3 – Bandwidth in the spectrum. In the options menu, change the value.

4 – List of scanned lists or bands.

5 – Options menu. Navigation through the menu **<** and **>**. Change the values with buttons **1** and **3**.

6 – Changing the modes of the spectrum analyzer Scanlists / Bands.

7 - Save Settings. Save after configuring and selecting the lists and the options will be applied once enabled.

8 – Change the display screen between Classic spectrum, Smooth spectrum and SCAN.

Smooth Spectrum

Classic Spectrum

9 – Change of AM/FM/USB modulation.

0 – Switch to the history mode. Displays the RX frequencies, channels, code and time in s.

***** and **#** DYNSQL **Level setting** - RSSI spike threshold for instant peak capture. For weak and fuzzy signals, increase or disable.

L* and **L#** DYNSQL OFF This will fill the history, but won't stop.

SK1 – Frequency Lock (FL) then monitor and back to normal mode.

FL locks the frequency, opens listening on receive

Monitor, keeps the frequency locked and open listening

SK2 – Blacklist a frequency, it is visible in history with a **#**

Keys in history

<,> - Move to previous / next item, automatically set to frequency lock to the selected item

1 – Clears all history (use 8 to save).

2 – Saves the frequency to a free channel.

3 – Delete the entry.

4 – Clears all black listed entries (use 8 to save).

5 – Scans history entries.

6 – Launch close call

7 - Clears all non black listed entries (use 8 to save).

8 –Saves the history.

EXIT : go back to SCAN or SPECTRUM

Keys in band selection

<,> - Move to previous / next item

1,3 – Select a preset

4 – Activate or deactivate one band.

5 – Activate one band and deactivate any other

7 – Save current preset.

MENU - activate selected item and start scanning

EXIT : go back to SCAN or SPECTRUM

Keys in scanlist selection

<,> - Move to previous / next item

4 – Activate or deactivate one scanlist.

5 – Activate one scanlist and deactivate any other

MENU - activate selected item and start scanning

EXIT : go back to SCAN or SPECTRUM

PARAMETERS OF THE SPECTRUM ANALYZER

SPECTRUMDELAY – Pause time after signal detection (in ms) – if 0, immediately resumes scanning.

MAXLISTENTIME – Maximum time to listen to the signal (in seconds) – after that, the scan continues.

FSTART – Initial frequency of the scan range (in MHz).

FSTOP – End frequency of the scan range (in MHz).

STEP – Frequency scan step (in kHz): 0.01; 0.05; 0.1; 0.5; 1.0; 2.0; 5.0; 6.25; 10.0; 12.5; 20.0; 25.0; 50.0; 100.0.

LISTENBW – Bandwidth: **Wide** (25 kHz) / **Narrow** (12.5 kHz)/ **AIR** (8.33kHz).

MODULATION – Modulation type: **FM**/**AM**/**USB**.

POWERSAVE – Power saving mode: Off /200ms /500ms /1s /2s /5s.

KEY UNLOCKED – THE TIME BEFORE THE KEYPAD IS LOCKED.

NOIS LVL OFF – Noise level for disabling reception (above this – the signal is counted).

POPUPS – Popup display time (in ms) – 0 = disabled.

RECORD TRIG – An additional threshold for recording in history. The higher the threshold, the stronger the signal will be recorded.

SOUND BOOST – Amplify sound. Increase bass frequencies.

PTT – What to do when pressing PTT: LAST VFO /NINJA /LAST RECEIVED/AUTO ROGER .

- NINJA chooses a random frequency in the current scanlist.
- AUTO ROGER :Choose your Roger in vfo menu, select a frequency or a channel in vfo, in spectrum select ptt auto Roger in the menu, it will send a Roger beep every duration you select.

MONITOR SL – Includes the "MON" **priority list in all scan lists**.

RESET DEFAULT – Reset all spectrum parameters to factory settings.

DETAILED MONITORING IN SPECTRUM MODE VIA THE "MENU" BUTTON

Move through the low-level register menu with buttons **2** and **8**. Change values with the **<=** and **=> buttons**

LNAS – Low Noise Amplifier (LNA) – signal amplification stages in front of the mixer.

LNA – Gain of the first LNA stage – higher = stronger signal but more interference.

PGA – Programmable Gain Amplifier (after LNA). – Balance between sensitivity and distortion.

MIX – Mixer Gain (Frequency Mixing) – Higher = Better Dynamic Range but More Noise.

XTAL F MODE SELECT – Quartz filter (XTAL) operating mode. Filtering (reduces noise, but can distort).

OFF AF RX DE-EMP – Disable demphasis (pre-distortion correction) during reception.

GAIN AFTER FM DEMOD – Adjusts the reception volume.

RF TX DEVIATION – Transmission frequency deviation (Fm deviation) – affects the transmission bandwidth (higher = louder).

COMPRESS AF TX RATIO – Compression ratio of audio during transmission – compresses the dynamic range.

COMPRESS AF TX 0 DB – Compression level 0 db during transmission – the compression start point.

COMPRESS AF TX NOISE – Noise reduction in quiet areas.

MIC AGC DISABLE – Disable automatic microphone gain control (AGC).

AFC RANGE SELECT – Operating range of AFC (Automatic Frequency Tuning).

AFC DISABLE – Disable AFC.

AFC SPEED – AFC Speed.

OPERATION IN BROADCAST RADIO MODE "FM"

The firmware has a simplified FM radio. Use **the <=** and **=> buttons** to search for the nearest available broadcasting frequency in automatic mode. The ***** button will change the **Auto mode** to **manual** search.

Press key 1 to 6 to select the corresponding memory.

Long press key 1 to 6 to save a frequency to that memory.

Press 0 to toggle receiving the VFO frequency, RX MON displayed or removed.

MAIN MENU

THERE IS NO HIDDEN MENU IN THE FIRMWARE

1.	**Step**	The frequency step (in kHz), UP and Down buttons change the frequency to this value, and you can only set the frequency to half of this value. For CB - 5/10 kHz; SatCom, Avia - 5/25 kHz; The rest is 6.25/12.5 kHz.
2.	**SQL**	The level of noise cancellation (Squelch) is from 0 (always open) to 9 (very strict). Set for both bands (A and B)
3.	**Mode**	Demodulation mode, default - FM, AM and USB can only be used for listening
4.	**RxMode**	Sets how the high and low frequencies are used on the screen (MAIN ONLY - Always transmits and listens on the main frequency (MO), DUAL RX RESPOND - Listens to both frequencies, if the signal is received on the secondary frequency, it is fixed on it for a couple of seconds so that you can answer the call (DWR), CROSS BAND - always transmits on the primary frequency and listens on the secondary frequency (XB), MAIN TX DUAL RX - always transmits to the main, listens to both (DW).
5.	**RxDCS**	Setting Digital Coded Squelch, if enabled, the squelch will only open after receiving that subtone, applicable for FM modulation only.
6.	**RxCTCS**	Setting an analog subtone to reception (Continuous Tone Coded Squelch), if the function is enabled, the noise canceler will open only after receiving this subtone, applicable only for FM modulation.
7.	**TxDCS**	Setting the digital encoded digital subtone to transmit, if the function is enabled, the transceiver will transmit this tone, applicable only for FM modulation.
8.	**TxCTCS**	Setting the analog subtone to transmit, if the function is enabled, the transceiver will transmit this tone, applicable for FM modulation only.
9.	**TxOffs**	Transmitter frequency offset value in MHz, transmit/receive at different frequencies.
10.	**TxODir**	Direction (+/-) of transmitter frequency offset when transmitting/receiving at different frequencies
11.	**W/N**	The bandwidth used by the transceiver (WIDE (25 kHz), NARROW (12.5 kHz).
12.	**SetNFM**	Sets Narrow FM to Narrow or Narrower
13.	**Scramb**	Selects Scrambling type
14.	**Scraen**	Enables Scrambler
15.	**Compnd**	Compander - Soft audio filter for clearer sound. Amplifies the received desired signal even when the signal is weak at the transmitting microphone, improving the signal-to-noise ratio. To work, it must be turned on on both walkie-talkies.
16.	**Roger**	On/off "roger beep" (end-of-pass tone).
17.	**STE**	Noise suppressor, eliminates noise at the end of the transmission. When PTT is released, a short 55Hz tone is emitted, signaling to other transceivers to mute
18.	**RP STE**	Noise Cancelling Tail Eliminator Repeater

19.	**ChSave**	Saves the frequency to a chosen memory
20.	**ChDele**	Deletes a channel memory
21.	**ChName**	Name of the memory channel.
22.	**ChDisp**	Choose how the saved channel will be displayed on the walkie-talkie display
23.	**F1Shrt**	Function assignment by short pressing the F1 side key.
24.	**F1Long**	Function assignment when the F1 side key is pressed for a long time.
25.	**F2Shrt**	Function assignment by short pressing the F2 side key.
26.	**F2Long**	Assign a function when you press the F2 side key for a long time.
27.	**M Long**	Assign a function when you long press the M key
28.	**KeyLck**	Automatically lock the keypad after a set time
29.	**TxOut**	Transmit Timeout - The amount of time that PTT can be held during transmission (in seconds). Protection against accidental clamping of PTT.
30.	**BatSav**	Battery Save: OFF / 1:1 / 1:2 / 1:4 / 1:8.
31.	**Mic**	Microphone sensitivity (microphone amplifier).
32.	**POnMsg**	Enable/disable the display of a message when turned on (Power On Message).
33.	**BLTime**	Screen backlight time when buttons are pressed.
34.	**BLMin**	Minimum backlight brightness (in percentage).
35.	**BLMax**	Maximum backlight brightness (in percentage).
36.	**BLTxRx**	Transmit/Receive Backlight: OFF/TX/RX/TX+RX.
37.	**SetInv**	Enable walkie-talkie screen color inversion
38.	**BtnInv**	Reverse the direction of the UP and DOWN buttons
39.	**SetLck**	Setting the keypad lock type with or without PTT.
40.	**BatTxt**	Battery Display Type, %, V or disabled.
41.	**MicBar**	Enabling/disabling the microphone bar
42.	**SetTmr**	Set the display of RX/TX timers on the screen
43.	**F Lock**	Setting up Frequency Lock. What frequencies are allowed or prohibited to work on.
44.	**BatCal**	Battery Calibration.
45.	**BatTyp**	Choice of battery type (Li-ion, NiMH, etc.) and capacity.
46.	**Reset**	Channel Preservation Reset (NOCH) or Full Factory Reset including Saved Channels (ALL)

Installation

RS232 version uses the kenwood cable and has 200 history entries
USB version uses USB-C cable and has 50 history entries because the USB driver takes more RAM memory.

RS232 Kenwood cable

USB-C cable

use the same procedure as F4HWN, after that :

- If necessary, you can do a full reset of the radio. press **the PTT** and **Fn2 buttons** , turn on the radio. A warning message will be displayed on the screen. hold the buttons for 4 seconds before resetting.

In case of issues to install, try to first install N7SIX firmware to reset the memory.

CHIRP Driver

Chirp_Sonic.py in DOWNLOAD_HERE

This driver also allows

- customization of the 50 bands on channels 976 to 1024.
  Select start frequency and offset as stop frequency, step and modulation

Chirp Configuration

- ➢ Memory channels 975 to 1024 store custom band settings. For each, define:
  - o Band Name.
  - o Start and End frequency (via offset).
  - o Modulation & Frequency Step.

	Frequency	Name	Tone Mode	Tone	Tone Squelch	DTCS	RX DTCS	DTCS Polarity	Duplex	Offset	Cross Mode	Mode	Tuning Step	Skip	Power	Busy Ch Lockout (Busycl)	Reverse Frequencies (R)	PTT ID (PTT ID)	Compander (Compnd)	Scanlists (Slist)
975	118.000000	AIR8.33k							+	137.000000		AM	8.33		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
976	144.000000	HAM144M							+	146.000000		FM	12.5		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
977	430.000000	HAM430M							+	440.000000		FM	10		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
978	446.000000	PMR446							+	446.200000		FM	6.25		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
979	225.000000	AIRMIL2							+	241.075000		AM	25		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
980	335.400000	AIRMIL3							+	339.700000		AM	25		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
981	146.000000	DMRVHF							+	170.000000		FM	12.5		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
982	450.000000	DMRUHF							+	470.000000		FM	12.5		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
983	155.500000	MARINE							+	162.025000		FM	25		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
984	26.515000	CB							+	28.305000		AM	5		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
985	433.075000	LPD433							+	433.775000		FM	6.25		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
986	14.000000	14MHz							+	14.300000		AM	5		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
987	17.400000	17MHz							+	17.800000		AM	5		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
988	18.068000	18MHz							+	18.168000		AM	1		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
989	21.000000	21MHz							+	21.450000		AM	1		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
990	24.890000	24MHz							+	24.990000		AM	1		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
991	50.000000	HAM50MHz							+	52.000000		FM	10		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
992	235.000000	SATCOM							+	275.000000		FM	10		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
993	14.000000	Full							+	1100.000000		FM	500		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
994	146.000000	146M							+	146.500000		FM	12.5		HIGH = 5W	Disabled	Disabled	OFF	OFF	OFF
995																				
996																				
997																				

- ➢ Scanlist Monitor setting Select up to 20ch to Monitor scanlist anywhere in the channels.

	Frequency	Name	Tone Mode	Tone	Tone Squelch	DTCS	RX DTCS	DTCS Polarity	Duplex	Offset	Cross Mode	Mode	Tuning Step	Skip	Power	Busy Ch Lockout (Busycl)	Reverse Frequencies (R)	PTT ID (PTT ID)	Compander (Compnd)	Scanlists (Slist)
901	446.006250	MON PMR1										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
902	446.018750	MON PMR2										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
903	446.031250	MON PMR3										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
904	446.043750	MON PMR4										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
905	446.056250	MON PMR5										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
906	446.068750	MON PMR6										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
907	446.081250	MON PMR7										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
908	446.093750	MON PMR8										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
909	446.106250	MON PMR9										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
910	446.118750	MON PMR10										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
911	446.131250	MON PMR11										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
912	446.143750	MON PMR12										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
913	446.156250	MON PMR13										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
914	446.168750	MON PMR14										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
915	446.181250	MON PMR15										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
916	446.193750	MON PMR16										NFM	6.25		LOW 1 = < 20mW	Disabled	Disabled	OFF	OFF	Monitor
917																				